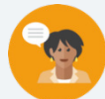


2.8 Adding, Subtracting, and Multiplying Polynomials Using More Than One Operation

Lesson Introduction

 Benchmark	MA.912.AR.1.3 Add, subtract, and multiply polynomial expressions with rational number coefficients.		 Learning Targets	- I can recognize adding, subtracting, or multiplying a polynomial will result in a polynomial. - I can add, subtract, and multiply polynomial expressions using more than one operation.	
 Benchmark Coherence	Building On		Working Toward		
	MA.8.AR.1.2 Apply properties of operations to multiply two linear expressions with rational coefficients.		MA.912.AR.1.5 Divide polynomial expressions using long division, synthetic division, or algebraic manipulation. MA.912.AR.1.9 Apply previous understanding of rational number operations to add, subtract, multiply, and divide rational algebraic expressions.		
 Essential Questions	- What role does the order of operations play in operations with polynomials? - How can I apply multiple operations in a single polynomial? - When adding, subtracting, or multiplying polynomials, will the result also be a polynomial?				
 Lesson Narrative	In this lesson, students apply what they have learned in previous lessons to perform multiple operations with polynomials. Students will begin by observing the connection between the order of operations and applying multiple operations to polynomials. They will then engage in a polynomial scavenger hunt. Students will draw conclusions that adding, subtracting, and/or multiplying polynomials will always result in a polynomial.				
 Component Summary	2.8.1 Warm-Up (~5 min.)	Career Video: Electric Vehicle Technician (<i>SE p. 97</i>)	2.8.3 Activity (20 min.)	Polynomial operations Scavenger Hunt (<i>SE p. 98</i>)	
	2.8.2 Collaboration (10-15 min.)	Connect the order of operations to performing operations with polynomials (<i>SE p. 97</i>)	2.8.4 Wrap-Up (~5 min)	Growth mindset activity (<i>SE p. 100</i>)	
 Resources	Glossary	Materials	Additional Preparation		
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A	- Polynomial Scavenger Hunt Cards (Optional) Calculators (Optional) Career Profile Video	Print, cut out, and hang the scavenger hunt images around the room to prepare for 2.8.3 Activity.		

 Teacher Tips	Common Misconceptions - Students may not correctly apply the order of operations with polynomial operations. - Students may not distribute to all terms inside the parentheses. - Students may combine terms in a polynomial that are not like terms.	Things to Consider There are a substantial number of cards that are available for the scavenger hunt in 2.8.3 Activity. Prior to instruction, consider sorting the cards into categories of difficulty to help you determine which cards students may need additional support with during the activity. This sorting can also be helpful in determining which cards students should start at.
	Literacy and Language Routines Take Turns In 2.8.3 Activity, students will work together to complete a polynomial scavenger hunt. They will take turns in the work of the activity. If they disagree, they are expected to support their case and listen to their partners' arguments.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.4.1: Discussion and Reflection 2.8.2 Collaboration and 2.8.3 Activity have students working together in partners or small groups. During these activities, students will communicate mathematical ideas and methods, as well as analyze the mathematical thinking of others in their group.
	 Differentiate Instruction	
Lesson Notes:		

2.8.1 Warm-Up: Electric Vehicle Technician

Component Overview: Students will engage in a discussion around improving something and having clear communication.



2.8.1 Warm-Up: Electric Vehicle Technician

Ben Grandy went from studying landscape architecture to becoming an electric vehicle technician because of his passion for fighting climate change.

Ben shared that he believes communication is a very important skill, regardless of your job title in a company. Asking questions to have clarity on product needs, being able to improve products, and always striving to achieve better helps you to grow.

1. Describe a time when you sought to improve in something.

Answers will vary. I tried to improve my batting average in baseball. I tried to improve my guitar playing to get better.

2. Describe a time when clear communication helped you improve.

Answers will vary. My math teacher was very clear with me on a project and I was able to complete all the steps and get a good grade. My football coach was clear in communicating specific plays and where I could improve, and I was able to get better.



This activity is accompanied by a video for this career. Consider having the video loaded up and ready to play when students enter the classroom to avoid technical issues.

2.8.2 Collaboration: Adding and Subtracting Polynomials With Distribution

Component Overview: Students will collaborate with a partner or small group to explore performing operations with polynomials using the order of operations when multiple operations are involved.



2.8.2 Collaboration: Adding and Subtracting Polynomials with Distribution

When performing operations on polynomials, the order of operations applies.

Work with your partner to complete the following.

1. The order of operations can be applied to simplify the polynomial $8x + 3(11x - 2)$.
 - a. Rewrite the expression in each step using the description.

	$8x + 3(11x - 2)$	Description
Step 1	$8x + 33x - 6$	Distribute 3 to $(11x - 2)$.
Step 2	$41x - 6$	Combine like terms.

- b. Complete the statement.

☐ Addition and subtraction
☒ **Multiplication**

is done before

☒ **addition and subtraction**
☐ multiplication

in the order of

operations.



For 2.8.2 Collaboration, provide the narrative instructions at the beginning of this lesson and then release students to work with their partner to complete the remainder of the activity prior to coming back together as a whole group to review their answers.



Why is multiplication done before addition and subtraction? What would happen if they were done in a different order?

2.8.2 Collaboration: Adding and Subtracting Polynomials With Distribution (continued)

2. Use the order of operations to simplify the polynomial and describe each step you perform.

$10\left(\frac{2}{5}m - 3p^2\right) + 2m - (7p^2 + 6)$	Description
$4m - 30p^2 + 2m - (7p^2 + 6)$	Distribute the 10
$4m - 30p^2 + 2m - 7p^2 - 6$	Distribute the -1
$6m - 30p^2 - 7p^2 - 6$	Combine the like terms with the variable 'm'
$-37p^2 + 6m - 6$	Combine the like terms with the 'p ² '

3. Complete the statement.

When adding and subtracting polynomials that include

distribution, first

- distribute monomials,
- add or subtract like terms,

then

- distribute monomials.
- add or subtract like terms.



Collect observational data as students are working through question 2. In particular, look to see whether students distribute the subtraction since it has been a few lessons since they subtracted polynomials. If a significant number of partners make this or another common error, consider facilitating a whole-group discussion of this problem to ensure understanding before moving on to the scavenger hunt in 2.8.3 Activity.



During whole-group review of question 2, consider a silent board game. Students volunteer to complete one box of the table that is posted at the front of the room, without talking. This strategy provides different entry points, as students choose the box for which they feel confident giving a response.

2.8.3 Activity: Polynomial Scavenger Hunt

Component Overview: Students will work with a partner or small group to complete a polynomial scavenger hunt. Students will continue the card scavenger hunt until they have completed 5 questions.



2.8.3 Activity: Polynomial Scavenger Hunt

1. With your partner, complete the polynomial scavenger hunt.
 - Start at one of the Polynomial Cards posted.
 - Draw (or describe) your card's emoji and write the polynomial expression on the recording sheet.
 - Simplify the expression by applying the order of operations.
 - Use your simplified answer to find your next card posted somewhere in the room.
 - Record the emoji on that Polynomial Card and your next expression to simplify.
 - Repeat until you have completed 5 problems.



For this activity, materials will need to be prepared and ready prior to the launch of this lesson. Print the pages and cut them in half to separate the cards, then hang them around the room.

Students will start at different cards so every student will not work every problem. Cards can be saved to be completed later during a unit review, possibly as a station, or as a challenge when students finish their work early.

2.8.3 Activity: Polynomial Scavenger Hunt (continued)

Polynomial Emoji Scavenger Hunt Recording Sheet	
Card Emoji	Polynomial Expression
	<i>Answers will vary based on the cards selected. Answers for each polynomial are located at the top of each card.</i> Answer:
	Answer:
	Answer:
	Answer:
	Answer:

The types of expressions you and your partner simplified in the emoji scavenger hunt were all polynomial expressions.

- Review your answers. Discuss with your partner what the results have in common.

Answers will vary.

- Complete the statement.

The result of adding, subtracting, and/or multiplying

polynomials will ☐ always ☐ not always result in a polynomial.



Avoid breezing over question 3. This is the basis for the concept of closure, which is key for students to understand conceptually prior to moving into dividing polynomials in Lesson 9.

2.8.4 Wrap-Up: Growth Mindset

Component Overview: Students will complete growth mindset self-reflection questions. This component should be done independently to determine how each student is engaging with a growth mindset.



2.8.4 Wrap-Up: Growth Mindset

1. Check the description that matches your answer to each question for today's lesson.

Answers will vary.

Mindset Question	Best Description for Me Today
Did I use the partner discussions as learning opportunities?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I work on problems that challenged me?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I use strategies to 'un-stick' myself when I found the problems difficult?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I put as much effort as I possibly could into the problems?	☐ Never ☐ Sometimes ☐ Usually ☐ Always



Consider pairing students to discuss their reflections to see if peers can provide each other support and insight into the strategies they can use to be successful when problems are challenging.